5 The soccer ball approaches a player at velocity $u = 12\ \text{m s}^{-1}$. Assume that the mass of the foot is much greater than that of the ball and that the collision is elastic.

At what velocity v should the player's foot move in order to stop the ball upon contact?

A $v = 12\ \text{m s}^{-1}$ in the direction opposite to the original velocity of the ball.

B $v = 6.0\ \text{m s}^{-1}$ in the direction opposite to the original velocity of the ball.

C $v = 6.0\ \text{m s}^{-1}$ in the same direction as the original velocity of the ball.

D $v = 0.12\ \text{m s}^{-1}$ in the same direction as the original velocity of the ball.